# Yusuke Ozaki

Kwansei Gakuin University, School of Engineering, Program of Computer Science
Former Exchange Student, Department of Computer Science, University at Albany, SUNY
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## OBJECTIVE

Seeking research opportunities in machine learning and AI for science.

Interested in problems at the intersection of Bayesian optimization, graph neural networks, active learning, and lab-automation–oriented machine learning.

Aiming to contribute to robust, data-efficient machine learning systems and tools, including optimization engines, recommender systems, and agent-based platforms for real-world applications.

## EDUCATION

**Kwansei Gakuin University | Apr. 2022 - Mar. 2027 (expected)**

Bachelor of Engineering in Computer Science, School of Engineering | Sanda, Hyogo, Japan

- GPA: 3.62/4.00

**University at Albany, SUNY | Aug. 2025 - May 2026**

Exchange Student, Department of Computer Science | Albany, NY, USA

**Kwansei Gakuin Senior High School | Apr. 2019 - Mar. 2022**

Graduated in March 2022, Japan.

## RESEARCH INTERESTS

AI for bioengineering and biology, including graph neural networks (GNNs) for molecular and cellular systems and gene regulatory network (GRN) inference from single-cell omics data.

Design of active-learning (Bayesian optimization) frameworks that autonomously drive scientific experiments, with a focus on protocol-aware lab automation and data-efficient exploration of experimental conditions.

Lipschitz bandits and structured planning with large language models.

## PUBLICATIONS

[1] Yusuke Ozaki and Dhaval Patel (2026). SPIN: Structural LLM Planning via Iterative Navigation for Industrial Tasks. arXiv:2605.14051 [cs.AI], submitted May 13, 2026. Preprint. https://arxiv.org/abs/2605.14051

## PRESENTATIONS

[1] Yusuke Ozaki (Sep. 2025). Object-Flow Machine Learning: Active learning framework utilizing protocols information. Poster presented in the Non-Archival Content Track at the AutoML Conference 2025 (AutoML 2025) , New York City, USA, Sep. 2025.

[2] Yusuke Ozaki (May 2025). Development of a machine learning method utilizing experimental protocols to predict outcomes and active learning framework employing the method. Oral presentation at the Student Symposium 2025, RIKEN Center for Biosystems Dynamics Research , Kobe, Japan, May 2025.

[3] Yusuke Ozaki (Feb. 2025). Development of a machine learning method utilizing protocols to predict experimental outcomes. Poster presentation at the Spring School for Theoretical Biology 2025 ( 2025) , Hiroshima University, Hiroshima, Japan, Feb. 2025.

## EXPERIENCE

**RIKEN Center for Biosystems Dynamics Research, Laboratory for Biologically Inspired Computing | Jun. 16, 2026 - Present**

Research Part-timer II | Japan

- Research team led by Koichi Takahashi. Current appointment began on June 16, 2026.

**RIKEN Center for Biosystems Dynamics Research, Laboratory for Biologically Inspired Computing Jun 2024 – Aug 2025**

Research Part-timer II Kobe, Japan

- Worked in Koichi Takahashi's Laboratory for Biologically Inspired Computing on AI- and robotics-driven automation of wet-lab experiments.
- Developed machine-learning based experimental simulators and protocol-aware active learning workflows for optimizing high-throughput experiments using automated liquid-handling systems.
- Contributed to Python codebases and data-processing pipelines for integrating experiment logs, protocol representations, and optimization results.

**Yachie Laboratory, Premium Research Institute for Human Metaverse Medicine (PRIMe), The University of Osaka Feb 2025 – Aug 2025**

Research Assistant | Osaka, Japan; laboratory led by Nozomu Yachie

- Assisted in computational projects on gene regulatory network inference from single-cell omics data, combining transcriptomic and chromatin-accessibility measurements.
- Built and maintained analysis pipelines for single-cell RNA-seq and ATAC-seq using Python-based tools and graph/network modeling methods.
- Collaborated with experimental biologists to interpret network models and compare alternative GRN-inference workflows.

**Metadata Incorporated Apr 2024 – Jun 2024**

Software Engineer (Part-time), Web Application Development Remote, Japan

- Developed and maintained web applications for AI-driven enterprise services, focusing on backend API implementation and integration with existing systems.
- Implemented and tested RESTful endpoints, simple user-facing interfaces, and data-processing components in collaboration with senior engineers.

## PROJECTS

**Analysis of Lipschitz Bandit Problems | Feb. 2026 - Present**

Supervisor: Chong Liu, University at Albany, SUNY, USA.

- Analyzing bandit problems using Lipschitz continuity.

**Bayesian Optimization with Gaussian Process Panel Models (GPM) | Aug 2025 – Present**

Supervisor: Chong Liu, University at Albany, SUNY, USA.

Tools: Python, Gaussian Processes, Preference-based Bayesian Optimization, Linear/Nonlinear Models

- Studying preference-based Bayesian optimization where observations are modeled by a General Preference Embedding Model (GPM), jointly learning latent utility functions and structured observation models at University at Albany, SUNY.
- Analyzing regret bounds and identifiability issues when combining learned embeddings with Gaussian process models for decision making.

**Minimal-Edges Graph Neural Networks for Efficient Graph Learning | Aug 2024 – Present**

Tools: Python, PyTorch, PyTorch Geometric, Graph Neural Networks, Reinforcement Learning

- Designed a graph neural network architecture that learns task-specific sparse adjacency structures layer by layer to reduce the number of edges used in convolution at Kwansei Gakuin University.
- Implemented an actor–critic style edge-selection module (Adjacency Generator) that optimizes graph sparsification using policy gradient methods.

**SPIN and Asset Operation LLM Agents | Aug. 2025 - Present**

Research collaborator: Dhaval Patel, IBM, USA. SPIN preprint released in May 2026.

Department of Computer Science, University at Albany, SUNY

- Developing a methodology and software framework for Asset Operation LLM Agents, combining large language models with structured task planning, tool calling, and evaluation pipelines.
- Implementing core components such as agent prompts, trajectory logging, and automatic DAG/plan evaluation to support reproducible experimentation on industrial asset-operations benchmarks.

## LEADERSHIP EXPERIENCE

**Technical Lead, Campus Activity Recommender (HackRPI 2025) | Nov. 2025 HackRPI 2025, Rensselaer Polytechnic Institute**

- Led a small team to design and implement a campus club and event recommender web application during a 24-hour hackathon, coordinating backend design (LinUCB bandit engine, Flask APIs, database schema).

**Team Lead, Capstone Project in Computer Science | Started Aug. 2025**

Department of Computer Science, University at Albany, SUNY

- Team lead for a multi-student senior capstone software project.

## SKILLS

Programming Languages: Python, C, Java

Web Technologies & APIs: Flask, RESTful APIs, JSON, Google Calendar API, Gemini API

Database Systems: PostgreSQL (incl. pgvector/pg_trgm), SQLite

Data Science & Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, matplotlib

DevOps & Version Control: Git, GitHub, Docker, Docker Compose, Linux (Ubuntu/Debian), pyenv, virtualenv

Specialized Areas: Graph Neural Networks (GNNs), Bayesian Optimization, Active Learning, Preference-based BO (GPM), Reinforcement Learning, Single-cell omics analysis, Gene Regulatory Network (GRN) inference, Lab-automation / protocol-aware ML

Mathematical & Statistical Tools: Linear algebra, probability theory, optimization, Bayesian modeling, graphical models (pgmpy)

## HONORS AND AWARDS

**Keyence Foundation "Ganbare! Nihon no Daigakusei" Student Support Grant | Jul. 2024**

**Gyomu Super Japan Dream Foundation Study Abroad Scholarship | Aug. 2025 - Jul. 2026**

## ADDITIONAL INFORMATION

Languages: Japanese (Native), English (Working proficiency; TOEFL iBT 79)

Scientific Interests: AI for Science and lab automation, synthetic biology and bioinformatics, graph neural networks and Bayesian optimization, single-cell omics and gene regulatory networks, active learning and preference-based optimization

Hobbies: Brazilian Jiu-Jitsu, reading books

## REFERENCES

1. Chong Liu Assistant Professor, Department of Computer Science University at Albany, SUNY Relationship: Research advisor (Bayesian optimization, Lipschitz bandits)

2. Koichi Takahashi Team Director (Principal Investigator), Laboratory for Biologically Inspired Computing RIKEN Center for Biosystems Dynamics Research (BDR), Kobe / Osaka, Japan Relationship: Research supervisor at RIKEN (wet-lab automation and AI for Science/robotics projects)

3. Akihiro Inokuchi Professor, Program of Computer Science, School of Engineering Kwansei Gakuin University, Hyogo, Japan Relationship: Undergraduate research advisor (graph mining, graph neural networks, data science)